WP7 Solutions

3. $V(x,y) = \pi x^2 y$, $V(2,5)$

$V(2,5) = \pi(2)^2 5$
$= 20\pi$

7. $g(x,y) = 4 - x - y$; $c = 0, 4$

$g(x,y) = 0 \Rightarrow 4 - x - y = 0$
$4 = x + y$ · a line
$g(x,y) = 4 \Rightarrow 4 - x - y = 4$
$x + y = 0$ a line through the origin

21. $g(x,y) = \frac{x}{x+y}$, $c = -1, 0, 2$

$g(x,y) = -1 \Rightarrow \frac{x}{x+y} = -1$
$g(x,y) = 0 \Rightarrow \frac{x}{x+y} = 0$
$g(x,y) = 2 \Rightarrow \frac{x}{x+y} = 2$

49. $w(x,y,z) = x^2 + y^2 + z^2$; $c = 9$

$x^2 + y^2 + z^2 = 9$
sphere of radius 3

51. $w(x,y,z) = x^2 + y^2 - z^2$; $c = 4$

$x^2 + y^2 - z^2 = 4$
hyperboloid of one sheet

56. $E(x,y,z) = \frac{K}{\sqrt{x^2+y^2}}$ $E = 10$, $E = 100$, $K = 1$

$E = 10$: $10 = \frac{1}{\sqrt{x^2+y^2}}$
$E = 100$: $100 = \frac{1}{\sqrt{x^2+y^2}}$

57. T is inversley proportional to the square of the distance from a point P to the origin

$T(x,y) = \frac{K}{x^2+y^2}$

58. $P(1,2) = 50°$; $Q(3,4)$

from #57 $T(x,y) = \frac{K}{x^2+y^2}$

$50 = \frac{K}{1^2+2^2}$
$50 = \frac{K}{5}$
$250 = K$

$T(3,4) = \frac{250}{3^2+4^2}$
$= \frac{250}{9+16}$
$= \frac{250}{25}$
$T(3,4) = 10$.

59. $T = 40^\circ$, $T = 100^\circ$

from 57: $T(x,y) = \frac{K}{x^2+y^2}$

$40 = \frac{K}{x^2+y^2}$

$x^2+y^2 = K/40$ (a circle w/ radius $\sqrt{10K}/20$)

$100 = \frac{K}{x^2+y^2}$

$x^2+y^2 = K/100$ (a circle w/ radius $\sqrt{K}/10$)

61. $\lim_{(x,y)\to(1,2)} \frac{5x^2y}{x^2+y^2}$

$= \frac{5(1)^2(2)}{1^2+2^2}$

$= \frac{10}{1+4}$

$= 2$

86. $\lim_{(x,y)\to(0,0)} \frac{xy+y^3}{x^2+y^2}$

a) along the x-axis

$\lim_{(x,y)\to(0,0)} \frac{xy+y^3}{x^2+y^2}$

$= \lim_{x\to 0} \frac{x(0)+0^3}{x^2+0^2}$

$= \lim_{x\to 0} 0$

$= 0$

b) along the y-axis

$\lim_{(x,y)\to(0,0)} \frac{xy+y^3}{x^2+y^2}$

$= \lim_{y\to 0} \frac{0(y)+y^3}{0^2+y^2}$

$= \lim_{y\to 0} y^3/y^2$

$= \lim_{y\to 0} y$

$= 0$

c) along $y = 2x$

$\lim_{(x,y)\to(0,0)} \frac{xy+y^3}{x^2+y^2}$

$= \lim_{x\to 0} \frac{x(2x)+(2x)^3}{x^2+(2x)^2}$

$= \lim_{x\to 0} \frac{2x^2+8x^3}{x^2+4x^2}$

$= \lim_{x\to 0} \frac{2+8x}{1+4}$

$= \frac{2}{5}$

87. $\lim_{(x,y)\to(0,0)} \frac{xy+y^3}{x^2+y^2}$

this limit is DNE since along the path $y=2x$ the limit is different than the limits along the x and y axes.

88. $\lim_{(x,y)\to(0,0)} \frac{x^2y}{x^4+y^2}$

a) along the x axis

$\lim_{(x,y)\to(0,0)} \frac{x^2y}{x^4+y^2}$

$= \lim_{x\to 0} \frac{x^2(0)}{x^4+0}$

$= \lim_{x\to 0} 0$

$= 0.$

88 cont. $\lim_{(x,y)\to(0,0)} \frac{x^2y}{x^4+y^2}$

b) along the y-axis

$$\lim_{(x,y)\to(0,0)} \frac{x^2y}{x^4+y^2}$$
$$= \lim_{y\to 0} \frac{0^2y}{0^4+y^2}$$
$$= \lim_{y\to 0} 0$$
$$= 0$$

c) along the path $y=x^2$

$$\lim_{(x,y)\to(0,0)} \frac{x^2y}{x^4+y^2}$$
$$= \lim_{x\to 0} \frac{x^2(x^2)}{x^4+(x^2)^2}$$
$$= \lim_{x\to 0} \frac{x^4}{2x^4}$$
$$= \frac{1}{2}$$

89. $\lim_{(x,y)\to(0,0)} \frac{x^2y}{x^4+y^2}$

this limit DNE
since along the path $y=x^2$ the limit is different than the limits along the x and y axes.